5.3.25

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Question

Solve the following pair of linear equations

$$3x + 4y = 10 \tag{1}$$

$$2x - 2y = 2 \tag{2}$$

Solution

Given equations:

$$\begin{pmatrix} 3 & 4 \end{pmatrix} \mathbf{x} = 10 \tag{3}$$

$$\begin{pmatrix} 2 & -2 \end{pmatrix} \mathbf{x} = 2 \tag{4}$$

In augmented matrix form:

$$\left(\begin{array}{cc|c} 3 & 4 & 10 \\ 2 & -2 & 2 \end{array}\right) \tag{5}$$

Solution

$$R_2 \to R_2 - \tfrac{2}{3}R_1 \Rightarrow \left(\begin{array}{cc|c} 3 & 4 & 10 \\ 0 & -\frac{14}{3} & -\frac{14}{3} \end{array}\right) \tag{6}$$

$$R_2 \to -\tfrac{3}{14}R_2 \Rightarrow \left(\begin{array}{cc|c} 3 & 4 & 10 \\ 0 & 1 & 1 \end{array}\right) \tag{7}$$

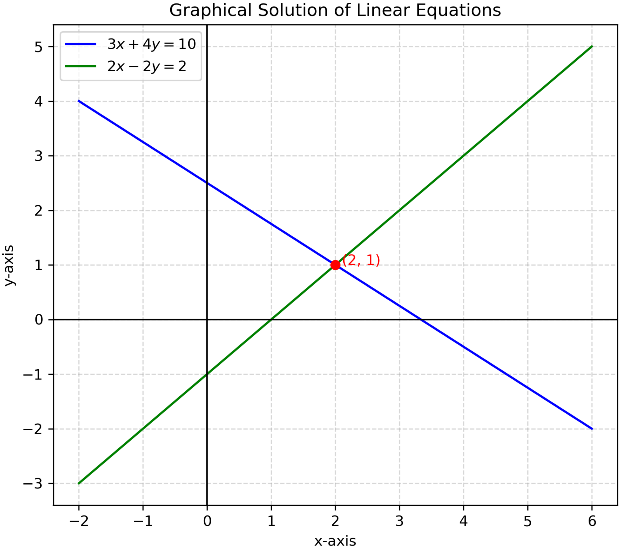
$$R_1 \to R_1 - 4R_2 \Rightarrow \left(\begin{array}{cc|c} 3 & 0 & 6 \\ 0 & 1 & 1 \end{array}\right) \tag{8}$$

$$R_1 \to \tfrac{1}{3}R_1 \Rightarrow \left(\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 1 \end{array}\right) \tag{9}$$

Hence,

$$x = 2, \quad y = 1. \tag{10}$$

Plot


Graphical Solution of Linear Equations
$3x + 4y = 10$
$2x - 2y = 2$
(2, 1)
y-axis
x-axis
5
4
3
2
1
0
−1
−2
−3
−2
−1
0
1
2
3
4
5
6

C Code

```
#ifndef LINEAR_H
#define LINEAR_H

#include <stdio.h>

// Function to take input for coefficients
void input_equations(float *a1, float *b1, float *c1, float *a
    printf("Enter coefficients of first equation (a1 b1 c1): "
    scanf("%f %f %f", a1, b1, c1);
    printf("Enter coefficients of second equation (a2 b2 c2):
    scanf("%f %f %f", a2, b2, c2);
}

// Function to compute determinant
float determinant(float a1, float b1, float a2, float b2) {
    return (a1 * b2) - (a2 * b1);
}
```

C Code

```
// Function to solve equations
void solve_equations(float a1, float b1, float c1, float a2, f
    float det = determinant(a1, b1, a2, b2);

    if (det == 0) {
        printf("The equations have no unique solution (paralle
    } else {
        float x = (c1 * b2 - c2 * b1) / det;
        float y = (a1 * c2 - a2 * c1) / det;
        printf("Solution: x = %.2f, y = %.2f\n", x, y);
    }
}

#endif
```

C Code

```
#include "solution.h"

int main() {
    float a1, b1, c1, a2, b2, c2;

    // Input the equations
    input_equations(&a1, &b1, &c1, &a2, &b2, &c2);

    // Solve the equations
    solve_equations(a1, b1, c1, a2, b2, c2);

    return 0;
}
```

Python Code

```
def determinant(a1, b1, a2, b2):
    """Compute the determinant of coefficient matrix"""
    return a1 * b2 - a2 * b1

def solve_equations(a1, b1, c1, a2, b2, c2):
    """Solve the two equations using Cramer's Rule"""
    det = determinant(a1, b1, a2, b2)

    if det == 0:
        print("The equations have no unique solution (paralle
    else:
        x = (c1 * b2 - c2 * b1) / det
        y = (a1 * c2 - a2 * c1) / det
        print(f"Solution: x = {x:.2f}, y = {y:.2f}")
```

Python Code

```python
def main():
    print("Enter coefficients for the equations:")
    a1, b1, c1 = map(float, input("a1 b1 c1: ").split())
    a2, b2, c2 = map(float, input("a2 b2 c2: ").split())

    solve_equations(a1, b1, c1, a2, b2, c2)

if __name__ == "__main__":
    main()
```

Python + C Code

```
import ctypes

# Load the shared library
lib = ctypes.CDLL('./solution.so')

# Define argument and return types for solve_equations
lib.solve_equations.argtypes = [ctypes.c_float, ctypes.c_float
                                  ctypes.c_float, ctypes.c_float
lib.solve_equations.restype = None

# Python wrapper function
def solve_from_c(a1, b1, c1, a2, b2, c2):
    lib.solve_equations(a1, b1, c1, a2, b2, c2)
```

Python + C Code

```python
# Take user input
print("Enter coefficients of first equation (a1 b1 c1): ")
a1, b1, c1 = map(float, input().split())

print("Enter coefficients of second equation (a2 b2 c2): ")
a2, b2, c2 = map(float, input().split())

# Call the C function
solve_from_c(a1, b1, c1, a2, b2, c2)
```